

Recap on Newton's 3 Laws of Motion

SS ILP Physics (Intermediate), Week 2 Class 4



Outline

- Recap on Newton's three Laws of Motion
- Questions and Applications

Newton's 1st Law of Motion

- An object remains its state of rest or uniform motion at a constant velocity unless it is acted upon a non-zero resultant force.

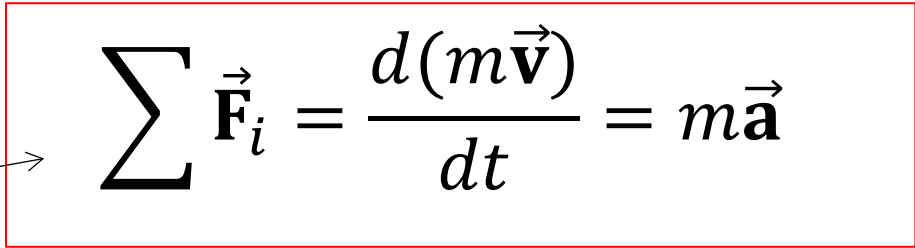
$$\sum \vec{F}_i = \vec{0} \rightarrow \vec{v} = \text{constant}$$

<https://www.physicsclassroom.com/class/newtlaws/Lesson-1/Newton-s-First-Law>

Newton's Second Law

- The acceleration is proportional to the resultant force, in the same direction and inversely proportional to the mass,

*Sum of all the forces
acting on that object


$$\sum \vec{F}_i = \frac{d(m\vec{v})}{dt} = m\vec{a}$$

<https://www.physicsclassroom.com/class/newtlaws/Lesson-3/Newton-s-Second-Law>

Newton's Third Law

- To every action, there is always an opposite reaction of the same magnitude,

$$\vec{F}_{2,1} = -\vec{F}_{1,2}$$

force on object 2 due to object 1

force on object 1 due to object 2

<https://www.physicsclassroom.com/class/newtlaws/Lesson-4/Newton-s-Third-Law>

Solving dynamics problems with Newton's three laws

- I. Identify the object(s) of interest.
- II. Draw free body diagram(s) of the object(s).
- III. Choose a proper coordinate system and determine the components of the forces and **acceleration** along coordinate axes.
- IV. Apply Newton's 2nd law of motion to write equations with unknowns,

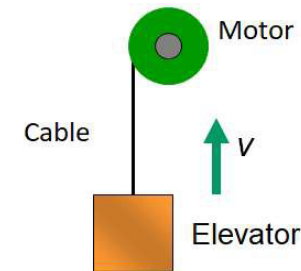
$$\sum_i^N \vec{F}_i = m\vec{a}$$

- V. In addition to Newton's 2nd law, identify any other equations you might need for solving unknowns.

H. D. Young and R. A. Freedman, University Physics (14th edition), p160.

Case problem 1 (conceptual)

- a. Consider a person standing in an elevator that is accelerating upward. Is the upward normal force N exerted by the elevator floor on the person smaller, larger or the same as the downwards force applied on the person? ([link](#))
- b. If the elevator is being lifted at a constant speed by a steel cable attached to an electric motor and we can neglect air resistance and any friction between the elevator and the surroundings, is the upwards force exerted on the elevator by the cable larger, smaller or the same as the gravitational attraction? ([link](#))



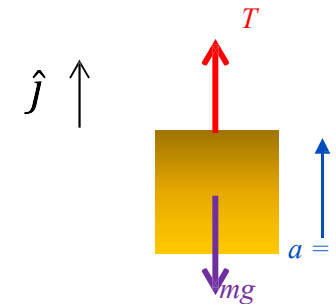
Case problem 1 - Solution

a. larger.

By Newton's 2nd law, the net force must be proportional to the acceleration and in the same direction. Since the person is accelerating upwards, the net force on the person must also be pointing upwards. The net force consists of the weight (downwards) and normal force from the floor (upwards), so the normal force must have a magnitude larger than the weight.

b. equal to the force of gravity.

Since the velocity is constant, the acceleration must be zero in the direction of motion. By Newton's 2nd law, the net force must be zero too. So, these two forces, tension and weight must have equal magnitude and opposite direction.



$$\begin{aligned} T\hat{j} + (-mg\hat{j}) &= ma\hat{j} \\ \Rightarrow T - mg &= ma \\ \Rightarrow T - mg &= 0 \\ \Rightarrow T &= mg \end{aligned}$$

Case problem 2

Consider the following system:



Both springs have a constant of $k = 25N/m$ and the block is motionless. If the bottom spring is compressed $0.4m$ past its equilibrium and the block has a mass of $3kg$, how far is the top spring stretched past its equilibrium? Consider $g = 10 m/s^2$ ([link](#))

Case problem 2 - Solution

Since the block is motionless, we know that our forces will cancel out: $F_{net} = 0$



There are three forces in play: one from each spring, as well as the force of gravity. If we assume that forces pointing up are positive, we can write:

$$F_{spring,top} + F_{spring,bot} - mg = 0$$

Plugging in expressions for each spring force, we get:

$$kx_{top} + kx_{bot} - mg = 0$$

Making x_{top} the subject, we get

$$x_{top} = \frac{mg - kx_{bot}}{k} = \frac{3kg \times 10 \frac{m}{s^2} - 25 \frac{N}{m} \times 0.4m}{25 \frac{N}{m}} = \frac{30N - 10N}{25 \frac{N}{m}} = 0.8m$$

Case problem 3

A 10kg weight is suspended by two ropes. Each rope makes an angle of 45 degrees to the horizontal.

What is the magnitude of the tension force in each rope? ([link](#))



$$g = 10 \frac{m}{s^2}$$

Case problem 3 - Solution



The fact that the angles of the two ropes are the same simplifies the problem. The tension in each rope will be the same. Now we just need to calculate what that force is.

We know that together, the vertical components of the tension must equal the weight of the block. Therefore, we can write:

$$T_1 \sin(45^\circ) + T_2 \sin(45^\circ) = mg$$

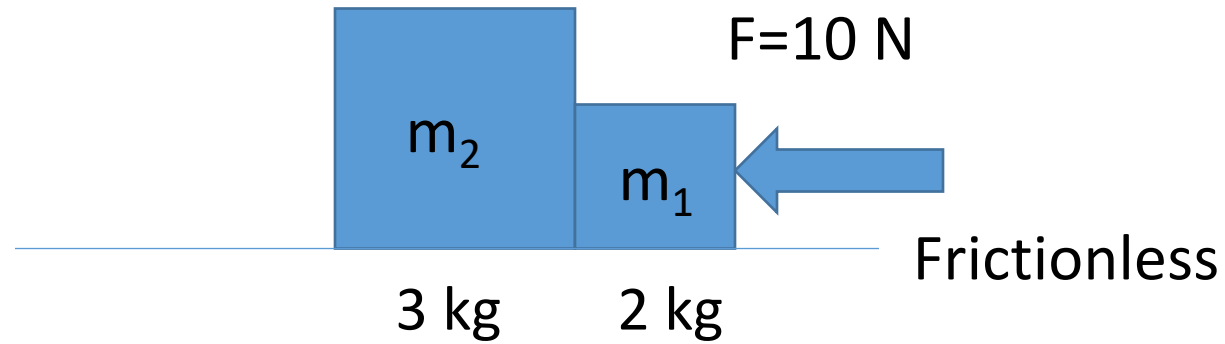
$$T_1 \sin(45^\circ) + T_2 \sin(45^\circ) = 100N$$

Since we know that the two tension forces are equal, we can rewrite: $2T \sin(45^\circ) = 100$

Rearranging for T , we get

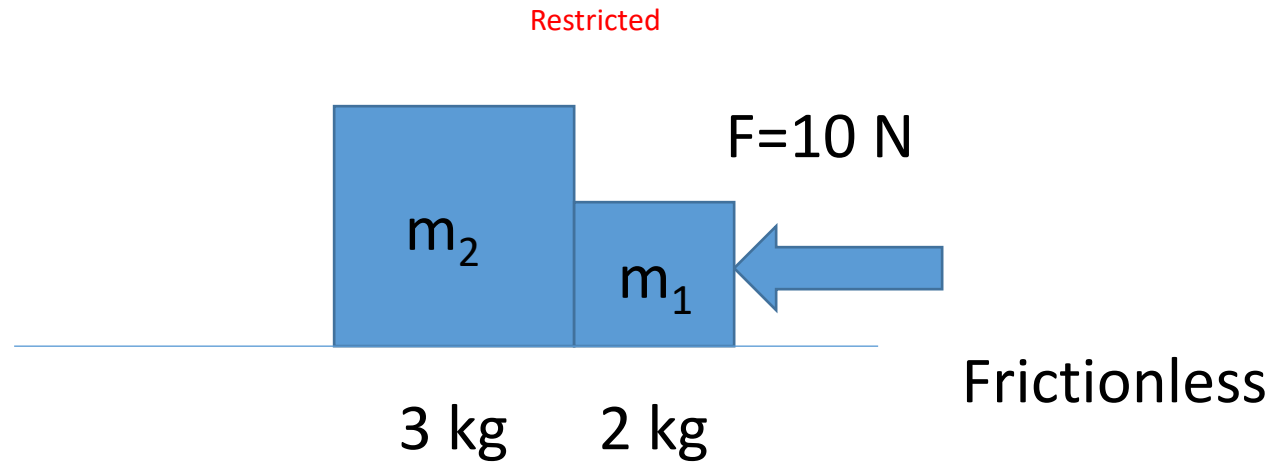
$$T = \frac{100}{2 \sin(45^\circ)} = 70.7N$$

Two objects moving with the same acceleration



What is the action and reaction force between the two boxes in x-direction?

Solution



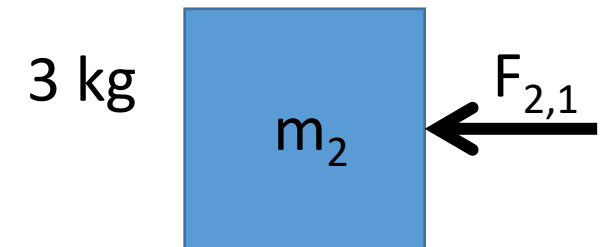
- Considering the two masses as a single system and applying Newton's 2nd Law,

$$\sum F = ma \rightarrow 10 = (m_1 + m_2)a$$

$$a = 2 \text{ m/s}^2$$

- Considering mass m_2 ,

$$\begin{aligned} \sum F = ma &\rightarrow F_{21} = m_2 a \\ F_{2,1} &= 6 \text{ N} = F_{1,2} \end{aligned}$$



Worked example 1 (conceptual)

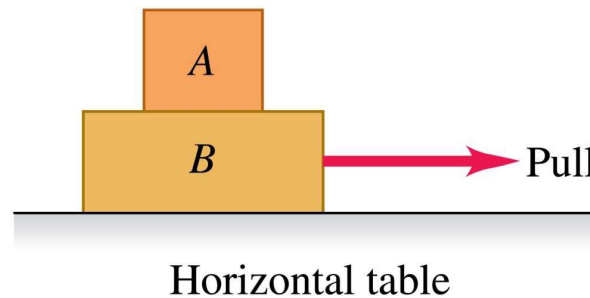
A person pulls horizontally on block B , causing both blocks to move horizontally as a unit. There is friction between block B and the horizontal table.

If the two blocks are moving to the right at constant velocity,

A. the horizontal force that B exerts on A points to the left.

B. the horizontal force that B exerts on A points to the right.

C. B exerts no horizontal force on A . ([link](#))



Worked example 1 - Solution

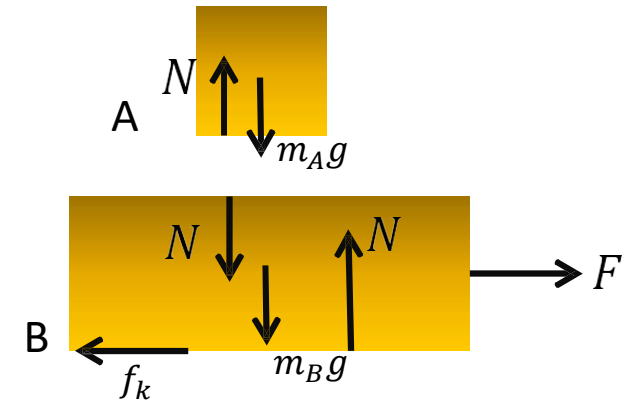
How do we start thinking about this problem?

The best way is to mentally separate the blocks and look at each one of them as a system subject to different forces of different origin (gravitational, friction, pull & push etc.)

Block A is only interacting with gravity and with block B through friction and a contact force. The weight of A, $m_A g$ is acting vertically and opposed to the reaction of B.

Constant velocity implies zero acceleration in the direction of motion. Thus, by Newton's 2nd law, the net force must also be zero. However, the only horizontal force that could be exerted on block A is the friction force at the interface due to block B (ignoring air resistance). Thus, block B must exert no friction force on block A.

So the answer is C.

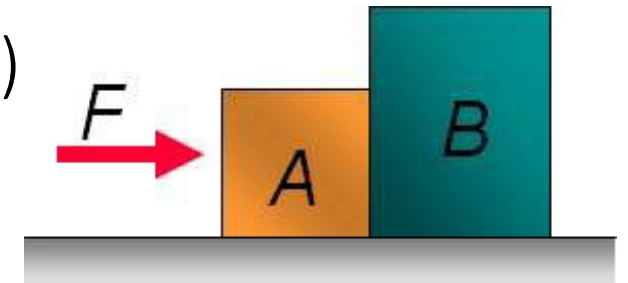


What if the air resistance is not negligible? How would the answer be different to maintain constant velocity for both the blocks?

Worked example 2 (conceptual)

A lightweight crate (A) and a heavy crate (B) are side-by-side on a horizontal floor. You apply a horizontal force F to crate A. There is friction between the crates and the floor. If the two crates are *accelerating to the right*,

- a. crate A exerts more force on crate B than B exerts on A
- b. crate A exerts less force on crate B than B exerts on A
- c. crate A exerts as much force on crate B as B exerts on A ([link](https://www.coursehero.com/tutors-problems/Physics/18879998-Please-refer-to-the-attachment-to-answer-this-question-This-question/))



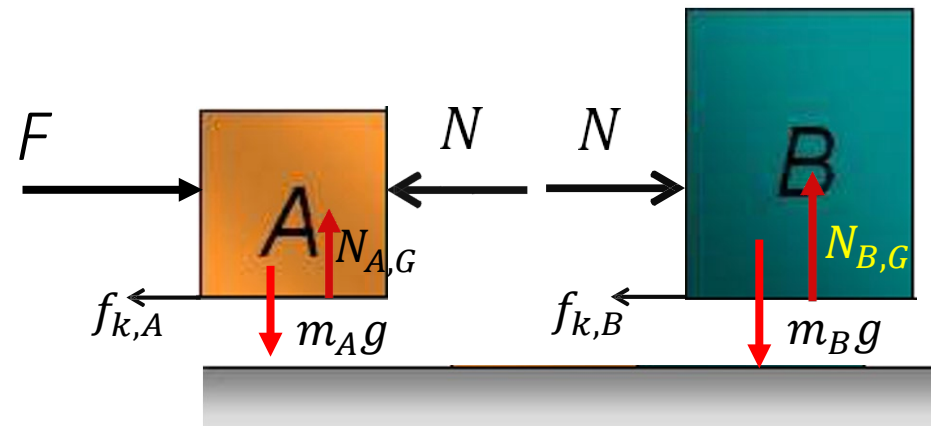
<https://www.coursehero.com/tutors-problems/Physics/18879998-Please-refer-to-the-attachment-to-answer-this-question-This-question/>

Worked example 2 Solution

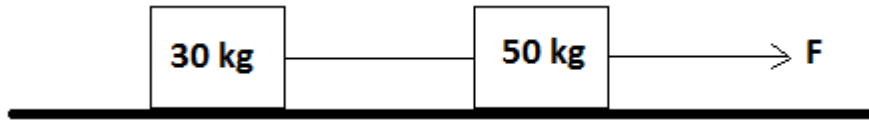
Both the crates accelerate to the right with the same acceleration (they are in contact with each other). Thus, the net forces on crate A and crate B are non-zero in the direction of motion (but they are not necessary the same, why?).

In the horizontal direction, crate A experiences the external force F , and also the normal contact force from B as well as friction from the floor. Crate B experiences only the normal contact force from A and friction. The two normal contact forces form the action and reaction pair by Newton's 3rd law of motion. So they are equal in magnitude and opposite in direction.

So the answer is C.



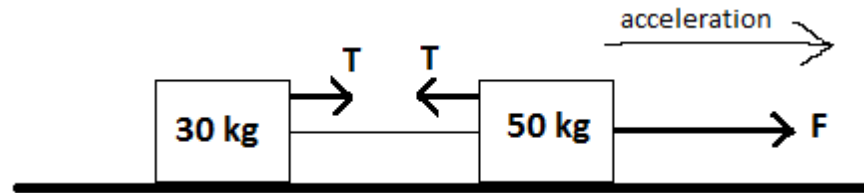
Case problem 4



In the diagram, a massless string connects two blocks of mass 30 kg and 50 kg that are on a flat, frictionless surface. A force F pulls on the 50 kg block, as shown. If the force F pulling on the 50 kg block is 100 N , what is the tension in the string connecting the two blocks?

Case problem 4 - Solution

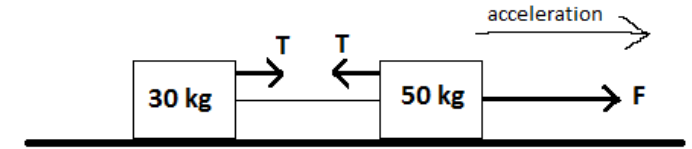
Start by drawing in the forces acting on each block. You could also draw in the force of gravity and the normal force for each block, but they have been omitted from the figure because they cancel each other out for each block and there is no friction to be considered here.



We are given in the question that the force F is 100 N . Since the blocks are connected by a string, they will accelerate at the same rate, and we can treat them as a system of total mass 80 kg (30 kg plus 50 kg). Use Newton's second law:

$$\sum \vec{F}_i = m\vec{a}$$

Case problem 4 - Solution



In this problem, the two tension forces form an action/reaction pair and therefore are equal in magnitude but opposite in direction (Newton's 3rd law). So:

$$100N - T + T = 80kg \cdot a$$

We can solve for acceleration, since the tensions cancel out.

$$a = \frac{100N}{80kg} = 1.25 \frac{m}{s^2}$$

Now that we have acceleration, we need to write a new equation in which the tension force does not cancel out so that we can solve for the tension T.

Do this by using Newton's second law again, except for only one of the blocks (the lighter one):

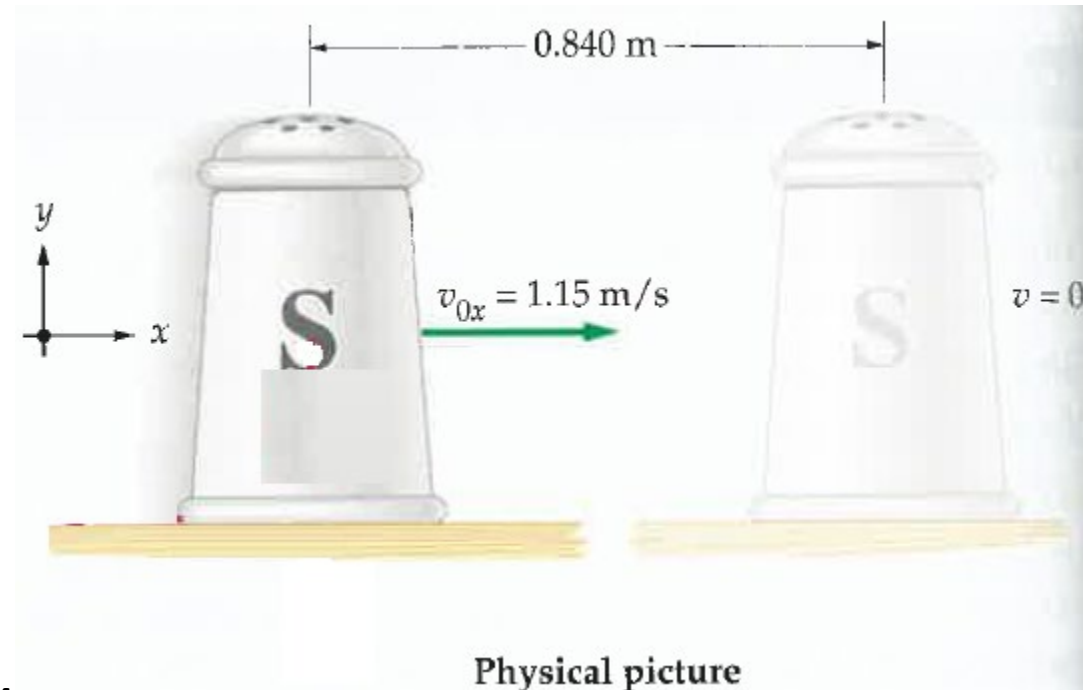
$$T = 30kg \cdot a = 30kg \cdot 1.25 \frac{m}{s^2} = 37.5N$$

Can you write a similar equation for the other block?

Off-class practice question(s)

Case problem

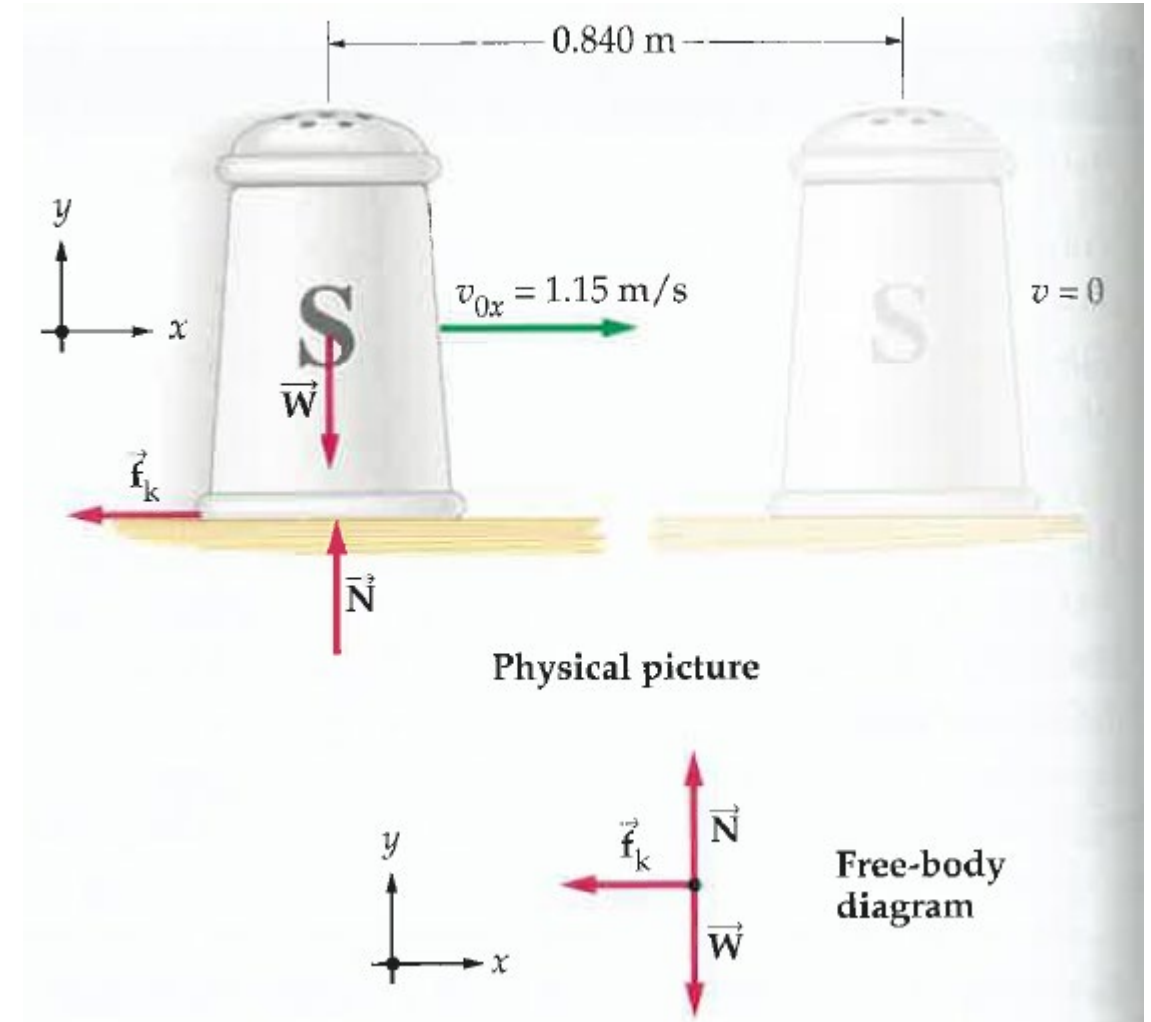
Someone at the other end of the table asks you to pass the salt. Feeling quite dashing, you slide the 50.0-g salt shaker in their direction, giving it an initial speed of 1.15 m/s. If the shaker comes to rest with constant acceleration in 0.840 m, what is the coefficient of kinetic friction between the shaker and the table. Take $g = 9.81 \text{ m/s}^2$.



J. S. Walker, Physics, Example 6-1, p144

Case problem: Solution

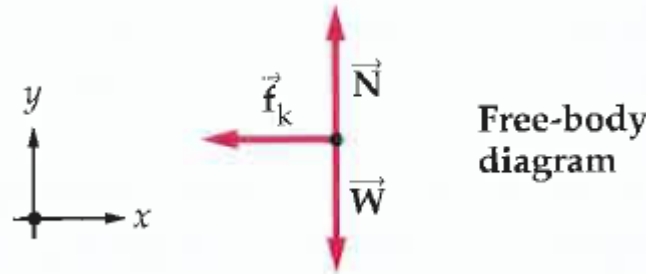
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Case problem: Solution

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$$\sum F_x = ma_x \rightarrow -f_k = -ma_x$$

$$\sum F_y = ma_y \rightarrow N - mg = 0 \rightarrow N = mg$$

J. S. Walker, Physics, Example 6-1, p144

$$v_x^2 - v_{x,0}^2 = -2a_x(x - x_0)$$

$$a_x = \frac{0 - 1.15^2}{-2(0.840 - 0)} = 0.787 \text{ m/s}^2$$

$$f_k = ma_x \rightarrow \mu_k N = ma_x$$

$$\mu_k = \frac{ma_x}{N} = 0.0802$$